

Course name: Databases

Laboratory work 1. ERD diagram.

Submission format: PDF document containing ER-Diagram of the system and textual description of the solution.

Required tasks.

For drawing ER-Diagram you must perform the following tasks:

1. **Identify Entities:** Make a list of all the entities mentioned in the system description.
2. **Determine Attributes:** For each entity, list down their attributes. Ensure to include primary keys, foreign keys, and any unique or required attributes, identify data types of each attribute.
3. **Identify Relationships:** Review the system description and determine the relationships between entities. Pay attention to the cardinality (one-to-one, one-to-many, many-to-many) and participation constraints.
4. **Draw Entity Boxes:** Use a diagramming tool to create a box for each entity identified. Label each box with the entity name.
5. **Add Attributes:** Inside each entity box, list the attributes identified for that entity. Include primary keys and any necessary foreign keys.
6. **Connect Entities with Relationships:** Draw lines between related entities to represent their relationships. Label the lines with the appropriate cardinality and participation constraints.
7. **Document the Diagram:** Provide a key or legend to explain any symbols or notations used in the diagram. Write a brief description or key explaining the meaning of each entity, attribute, and relationship.

International Railway Station

The Railway Station Database is designed to manage various aspects of railway operations including passenger bookings, trains, railway companies, tickets, and luggage details.

It's assumed that you've purchased a train ticket online at least once, so the general structure should be familiar.

An international railway system has a main table that stores data related to railway stations such as station id, station name, country, state, city. It also indicates when the station information was created and updated.

The system stores information about scheduled and actual train trips. Attributes include trip id, departure platform, arrival platform, when the trip was created and updated, railway company id, departure station id and arrival station id, scheduled departure time, scheduled arrival time, actual departure time, and actual arrival time.

Train trips are operated by different railway companies. Each railway company records information including company id, company code, company name, country, and when the company information was created and updated.

Each trip can have tickets purchased by passengers. Attributes include ticket id, trip id, ticket status, booking platform, when the ticket was created and updated, ticket price, and passenger id. If any changes are made to a passenger's ticket, that information should be stored in a separate table.

The system collects details about boarding tickets issued to passengers. This includes boarding pass id, seat number, boarding time, when the boarding pass was created and updated, and ticket id.

During the booking of a train trip, passengers can register luggage if they have any. For each piece of luggage, the system should keep luggage id, weight in kg, created date, updated date, and ticket id.

After registering luggage, each piece of luggage belonging to a ticket needs to be inspected at the railway station. The system collects information such as luggage inspection id, inspection result, created at, updated at, ticket id, and passenger id.

The system stores passenger information associated with tickets. Attributes include passenger id, first name, last name, gender, date of birth, country of citizenship, country of residence, passport number, and when the passenger profile was created and updated. Also, each passenger passes through railway station security before boarding the train.

The system collects information about the security check, including security check id, check result, when the information was created and updated, and passenger id.

Relationships and Constraints:

Various tables have primary keys and foreign key constraints to maintain data integrity and enforce relationships between entities.

For example, flights are associated with airports for departure and arrival, bookings are linked to accounts, passengers, and booking legs, and so on.

This database facilitates the management of airport-related operations, including passenger bookings, flight scheduling, aircraft information, and frequent flyer programs, ensuring efficient and organized airport services.

These tables establish various types of relationships, including:

One-to-Many: One record in one table corresponds to many records in another table.

Many-to-One: Many records in one table correspond to one record in another table. *One-to-One*: Each record in one table is related to exactly one record in another table, and vice

versa.

Many-to-Many (not explicitly shown in this schema): Many records in one table can be related to many records in another table through a linking table.

These relationships facilitate efficient data management and retrieval, ensuring that the database accurately represents the complex interactions within the airport system.

Laboratory Work 1: International Railway Station ERD System Description

1. Executive Summary

This report presents the database design and Entity-Relationship Diagram (ERD) for the International Railway Station Management System. The database is structured to handle key operational aspects, including passenger registration, ticket issuance, trip scheduling, railway company management, luggage tracking, security screening, and ticket audit logging.

2. Entity Descriptions & Key Attributes

Entity Name	Attribute Name	Data Type	Key / Constraint	Description
stations	station_id	serial	Primary Key	Уникальный идентификатор станции
	station_name	varchar(67)	Not Null	Название станции
	country	varchar(67)	Not Null	Страна
	state	varchar(67)	Optional	Область/ Штат
	city	varchar(67)	Not Null	Город
	created_at	timestamp	Default Current	Дата создания записи
	updated_at	timestamp	Default Current	Дата обновления записи
railway_companies	company_id	serial	Primary Key	Уникальный ID компании
	company_code	varchar(67)	Unique, Not Null	Код компании
	company_name	varchar(67)	Not Null	Название компании
	country	varchar(67)	Not Null	Страна регистрации
	created_at	timestamp	Default Current	Дата создания записи
	updated_at	timestamp	Default Current	Дата обновления записи
passengers	passenger_id	serial	Primary Key	Уникальный ID пассажира
	first_name	varchar(67)	Not Null	Имя
	last_name	varchar(67)	Not Null	Фамилия
	gender	varchar(67)	Optional	Пол

	date_of_birth	date	Not Null	Дата рождения
	country_of_citizenship	varchar(67)	Not Null	Гражданство
	country_of_residence	varchar(67)	Not Null	Страна проживания
	passport_number	varchar(67)	Unique, Not Null	Номер паспорта
	created_at	timestamp	Default Current	Дата создания профиля
	updated_at	timestamp	Default Current	Дата обновления профиля
trips	trip_id	serial	Primary Key	Уникальный ID рейса
	company_id	integer	Foreign Key	Ссылка на railway_companies
	departure_station_id	integer	Foreign Key	Ссылка на stations (отправление)
	arrival_station_id	integer	Foreign Key	Ссылка на stations (прибытие)
	departure_platform	varchar(67)	Optional	Платформа отправления
	arrival_platform	varchar(67)	Optional	Платформа прибытия
	scheduled_departure_time	timestamp	Not Null	Плановое время отправления
	scheduled_arrival_time	timestamp	Not Null	Плановое время прибытия
	actual_departure_time	timestamp	Optional	Фактическое время отправления
	actual_arrival_time	timestamp	Optional	Фактическое время прибытия
	created_at	timestamp	Default Current	Дата создания записи
	updated_at	timestamp	Default Current	Дата обновления записи
tickets	ticket_id	serial	Primary Key	Уникальный ID билета
	trip_id	integer	Foreign Key	Ссылка на trips
	passenger_id	integer	Foreign Key	Ссылка на passengers
	ticket_price	numeric(10,2)	Not Null	Стоимость билета
	ticket_status	varchar(67)	Not Null	Статус (confirmed, cancelled)
	booking_platform	varchar(67)	Optional	Платформа бронирования
	created_at	timestamp	Default Current	Дата покупки билета
	updated_at	timestamp	Default Current	Дата изменения билета
ticket_audit_logs	log_id	serial	Primary Key	Уникальный ID лога
	ticket_id	integer	Foreign Key	Ссылка на tickets

	previous_status	varchar(67)	Optional	Предыдущий статус билета
	new_status	varchar(67)	Optional	Новый статус билета
	changed_at	timestamp	Default Current	Время изменения
	change_description	text	Optional	Описание изменений
boarding_passes	boarding_pass_id	serial	Primary Key	Уникальный ID талона
	ticket_id	integer	Foreign Key, Unique	Ссылка на tickets (1:1)
	seat_number	varchar(67)	Not Null	Номер места
	boarding_time	timestamp	Not Null	Время посадки
	created_at	timestamp	Default Current	Дата выгрузки талона
	updated_at	timestamp	Default Current	Дата обновления
luggage	luggage_id	serial	Primary Key	Уникальный ID багажа
	ticket_id	integer	Foreign Key	Ссылка на tickets
	weight_kg	numeric(5,2)	Not Null	Вес багажа в кг
	created_at	timestamp	Default Current	Дата регистрации багажа
	updated_at	timestamp	Default Current	Дата изменения
luggage_inspections	inspection_id	serial	Primary Key	Уникальный ID досмотра
	luggage_id	integer	Foreign Key	Ссылка на luggage
	ticket_id	integer	Foreign Key	Ссылка на tickets
	passenger_id	integer	Foreign Key	Ссылка на passengers
	inspection_result	varchar(67)	Not Null	Результат (passed, failed)
	created_at	timestamp	Default Current	Время проведения досмотра
	updated_at	timestamp	Default Current	Дата обновления записи
security_checks	security_check_id	serial	Primary Key	Уникальный ID проверки
	passenger_id	integer	Foreign Key	Ссылка на passengers
	check_result	varchar(67)	Not Null	Результат (cleared, flagged)
	created_at	timestamp	Default Current	Время досмотра
	updated_at	timestamp	Default Current	Дата обновления записи

3. Relationship Summary & Constraints

Parent Table	Child Table	Relationship Type	Foreign Key Field	Description
railway_companies	trips	One-to-Many (1:N)	company_id	Одна компания осуществляет множество рейсов
stations	trips	One-to-Many (1:N)	departure_station_id, arrival_station_id	Станция служит местом отправления/прибытия
passengers	tickets	One-to-Many (1:N)	passenger_id	Один пассажир может купить несколько билетов
trips	tickets	One-to-Many (1:N)	trip_id	На один рейс продается множество билетов
tickets	boarding_passes	One-to-One (1:1)	ticket_id	Одному билету соответствует ровно один посадочный талон
tickets	ticket_audit_logs	One-to-Many (1:N)	ticket_id	Изменение билета записывает историю в лог
tickets	luggage	One-to-Many (1:N)	ticket_id	На один билет можно зарегистрировать несколько сумок
luggage	luggage_inspections	One-to-Many (1:N)	luggage_id	Каждый багаж проходит досмотр
passengers	security_checks	One-to-Many (1:N)	passenger_id	Пассажир проходит досмотр перед посадкой

4. Notation & Legend

Notation / Symbol	Name	Description
PK	Primary Key	A unique identifier for a row in a table. It ensures each record is unique.
FK	Foreign Key	A key that connects one table to another table. It maintains referential integrity between tables.
Crow's Foot Notation	Crow's Foot Notation	A standard visual style used to show relationships between entities in ER diagrams.

